PERCENTAGES, GEOMETRIC AVERAGES AND LOGARITHMS ON ONE PAGE

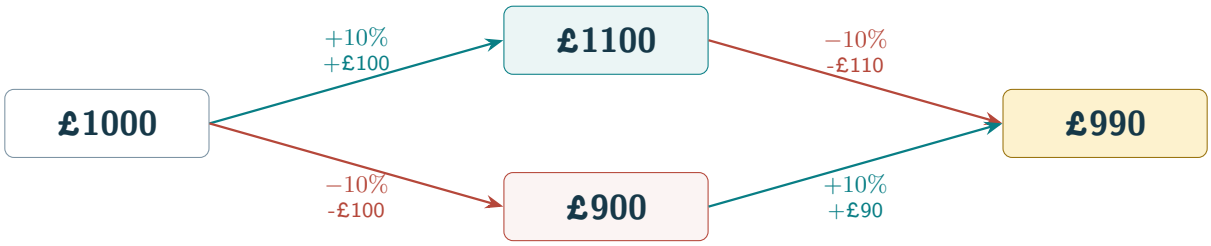
The rightful fear of loss

Why an equal percentage gain cannot undo a percentage loss.

1 / FOLLOW THE SAME MONEY

The percentage base changes at each step

Start with £1000. Each change is a percentage of the pot *at that moment*.



$$1.1 \times 0.9 = 0.99, \qquad (1+p)(1-p) = 1-p^2 \quad (0 < p < 1)$$

Either order loses 1% of the starting pot. In general, an equal up-and-down pair loses the fraction p^2; here $p = 0.10$.

2 / WHAT WOULD ACTUALLY UNDO THE LOSS?

The climb back is steeper

A loss of fraction p leaves $(1-p)P$. To recover the original P, the next gain q must satisfy

$$(1-p)(1+q) = 1.$$

$$q = \frac{1}{1-p} - 1 = \frac{p}{1-p}$$

loss	gain to recover
10%	11.11%
20%	25%
25%	33.33%
50%	100%
75%	300%
90%	900%

At 100% loss, every finite multiplier still gives zero.

3 / WHICH AVERAGE FOLLOWS THE POT?

Average the growth factors

The arithmetic average of $+10\%$ and -10% is 0%. The constant rate g giving the same two-period outcome instead solves $(1+g)^2 = 0.99$. Its return per period is

$$g = \sqrt{0.99} - 1 \approx -0.5013\%.$$

For n equal-length periods with fractional returns $r_1, \ldots, r_n$, take the geometric mean of their *multipliers*:

$$1 + g = \left(\prod_{k=1}^{n} (1 + r_k) \right)^{1/n}$$

Then $P_n = P_0(1+g)^n$. This geometric rate never exceeds the arithmetic mean of the same returns; equality holds only when all the period returns are identical.

4 / PRODUCTS BECOME SUMS

Logarithmic returns add exactly

A fractional return r_k multiplies the pot by $1 + r_k$. Define its **logarithmic return** by $L_k = \ln(1 + r_k)$. Then

$$\ln\left(\frac{P_n}{P_0}\right) = \sum_{k=1}^{n} L_k.$$

For our pair, $\ln(1.1) \approx 0.09531$ and $\ln(0.9) \approx -0.10536$. These add to

$$\ln(1.1) + \ln(0.9) = \ln(0.99) \approx -0.01005.$$

A true reversal uses *reciprocal* multipliers:

$$0.9 \times \frac{10}{9} = 1, \qquad \ln(0.9) + \ln\left(\frac{10}{9}\right) = 0$$

Average the log returns to get $\overline{L}$. Exponentiating gives the geometric rate: $g = \exp(\overline{L}) - 1$.

5 / THE GEOMETRY OF RECOVERY

Bigger losses demand much bigger gains

The solid curve gives the recovery gain; the dashed line shows an equal percentage gain.



As $p \to 1$, the recovery gain $p/(1-p)$ grows without bound.

Small shortfalls compound too. Fifty $+10\%, -10\%$ pairs leave $0.99^{50} \approx 0.605$ of the starting pot: about £605 from £1000.

Conventions. One pot, no payments in or out; $P_0 > 0$ and $r_k > -1$ for logarithms. Equal cash gains and losses cancel; equal percentage changes act on different bases.

Further reading. Will Morrison, *The Myth of Volatility Drag (Part 1)*, CFA Institute, 23 March 2015.

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